MODULE 4
ABSTRACT ALGEBRAIC SYSTEMS & LATTICES

Discrete Mathematical Structures

Important

Exam Focus:
The important areas of this module include:

- Binary operations and their properties
- Semigroup, monoid, group and abelian group
- Subgroups
- Homomorphism and isomorphism
- Lattice, meet and join
- Types of lattices
- Simple theorems related to groups

Contents

1 Algebraic Systems **4**
1.1 Binary Operation . . . 4
1.2 Examples . . . 5
1.3 Properties of Binary Operations . . . 5
1.3.1 1. Closure . . . 5
1.3.2 2. Associative Property . . . 6
1.3.3 3. Commutative Property . . . 6
1.3.4 4. Identity Element . . . 6
1.3.5 5. Inverse Element . . . 7
1.4 Algebraic System . . . 7

2 Semigroup, Monoid and Group **7**
2.1 Semigroup . . . 8
2.2 Monoid . . . 9
2.3 Group . . . 10
2.4 Abelian Group . . . 11

3 Subgroups **11**
3.1 Subgroup Test . . . 11
3.2 PYQ: Intersection of Two Subgroups . . . 12

4 Homomorphism **13**
4.1 How to Prove a Function is a Homomorphism 14
4.2 Example 14

5 Isomorphism **15**
5.1 PYQ: Homomorphism from $\mathbb{Z}$ to $\mathbb{Z}_4$ 15

6 Groups: Important Example **16**
6.1 Step 1: Closure 16
6.2 Step 2: Associativity 17
6.3 Step 3: Identity Element 18
6.4 Step 4: Inverse Element 18
6.5 Step 5: Conclusion 19

7 Lattices **19**
7.1 Partial Order 19
7.2 Lattice Definition 20
7.3 Understanding Lower Bound and Upper Bound 20
7.4 Example of a Lattice 20

8 Basic Laws of Lattices **21**
8.1 1. Idempotent Law 21
8.2 2. Commutative Law 22
8.3 3. Associative Law 22
8.4 4. Absorption Law 22

9 Types of Lattices **22**
9.1 1. Bounded Lattice 23
9.2 2. Complemented Lattice 23
9.3 3. Distributive Lattice 24
9.4 4. Complete Lattice 24
9.5 Quick Comparison of Types of Lattices 24

10 Important Theorems **24**
10.1 Theorem 1: Uniqueness of Identity 24
10.2 Theorem 2: Uniqueness of Inverse 25
10.3 Theorem 3: Inverse of a Product 25
10.4 Theorem 4: Subgroup of an Abelian Group 26

11 Previous-Year Questions: Quick Revision **27**

12 Exam Writing Strategy **27**
12.1 If Asked: Define Semigroup 27
12.2 If Asked: Prove $(\mathbb{Z}, +)$ is a Monoid 27
12.3 If Asked: Verify Whether an Operation Forms a Group 28
12.4 If Asked: Prove Homomorphism 28
12.5 If Asked: Define Lattice 28

13 One-Page Quick Revision **29**

14 Memory Shortcuts **30**

1. Algebraic Systems

1.1. Binary Operation

A **binary operation** is an operation that combines two elements of a set and produces another element of the same set.

Mathematically,

$$\boxed{* : A \times A \to A}$$

This means that for every $a, b \in A$,

$$\boxed{a * b \in A}$$

The important point is that the result must remain inside the set.

Definition

A binary operation on a non-empty set A is a function

$$* : A \times A \to A$$

which assigns an element of A to every ordered pair $(a, b) \in A \times A$.

1.2. Examples

Example

Example 1: Addition on integers
Let

$$A = \mathbb{Z}.$$

For any $a, b \in \mathbb{Z}$,

$$a + b \in \mathbb{Z}.$$

Therefore, addition is a binary operation on $\mathbb{Z}$.

Example 2: Multiplication on natural numbers
Let

$$A = \mathbb{N}.$$

For any $a, b \in \mathbb{N}$,

$$a \times b \in \mathbb{N}.$$

Therefore, multiplication is a binary operation on $\mathbb{N}$.

Example 3: Subtraction on natural numbers
Consider

$$2 - 5 = -3.$$

But

$$-3 \notin \mathbb{N}.$$

Therefore, subtraction is **not** a binary operation on $\mathbb{N}$.

1.3. Properties of Binary Operations

A binary operation may have several important properties.

1.3.1 1. Closure

An operation $*$ on A is closed if

$$\boxed{a * b \in A \quad \forall a, b \in A}$$

In simple words:

> Take any two elements from the set. Their result must also belong to the same set.

1.3.2 2. Associative Property

An operation is associative if

$$\boxed{(a * b) * c = a * (b * c)}$$

for all $a, b, c \in A$.
The grouping does not affect the answer.

Example

For addition,

$$(2 + 3) + 4 = 2 + (3 + 4)$$

$$5 + 4 = 2 + 7$$

$$9 = 9.$$

Therefore, addition is associative.

1.3.3 3. Commutative Property

An operation is commutative if

$$\boxed{a * b = b * a}$$

for all $a, b \in A$.
In simple words, changing the order does not change the result.

Example

For addition,

$$2 + 3 = 3 + 2 = 5.$$

Therefore, addition is commutative.

1.3.4 4. Identity Element

An element $e \in A$ is called an identity element if

$$\boxed{a * e = e * a = a}$$

for every $a \in A$.
Examples:

$$\text{Identity for addition} = 0$$

because

$$a + 0 = 0 + a = a.$$

The identity for multiplication is 1:

$$a \times 1 = 1 \times a = a.$$

1.3.5 5. Inverse Element

Suppose e is the identity element.

An element b is called the inverse of a if

$$\boxed{a * b = b * a = e}.$$

For addition:

$$a + (-a) = 0.$$

Therefore, the additive inverse of a is

$$\boxed{-a}.$$

For multiplication, the inverse of $a \neq 0$ is

$$\boxed{\frac{1}{a}}.$$

1.4. Algebraic System

An **algebraic system** consists of a non-empty set together with one or more operations defined on the set.

Examples:

$$(A, *)$$

and

$$(A, +, \times).$$

2. Semigroup, Monoid and Group

There is a useful sequence to remember:

$$\boxed{\text{Semigroup} \rightarrow \text{Monoid} \rightarrow \text{Group} \rightarrow \text{Abelian Group}}$$

Each step adds one more property.

Property	Semigroup	Monoid	Group	Abelian Group
Closure	Yes	Yes	Yes	Yes
Associativity	Yes	Yes	Yes	Yes
Identity	No requirement	Yes	Yes	Yes
Inverse	No requirement	No requirement	Yes	Yes
Commutativity	No requirement	No requirement	No requirement	Yes

Important

Memory Trick

$$\boxed{\text{C A I I C}}$$

Think:

- C = Closure
- A = Associativity
- I = Identity
- I = Inverse
- C = Commutativity

Semigroup: CA
Monoid: CA + I
Group: CA + II
Abelian Group: CA + IIC

2.1. Semigroup

Definition

A **semigroup** is a non-empty set together with a binary operation that is:

1. Closed
2. Associative

Therefore,

$$\boxed{\text{Semigroup} = \text{Closure} + \text{Associativity}}$$

Example

Consider

$$(\mathbb{N}, +).$$

For $a, b \in \mathbb{N}$,

$$a + b \in \mathbb{N}.$$

So addition is closed.
Also,

$$(a + b) + c = a + (b + c).$$

Therefore, $(\mathbb{N}, +)$ is a semigroup.

Previous-Year Question

Question: Define semigroup with example.
Answer:
A semigroup is a non-empty set together with a binary operation which is closed and associative.
For example,

$$(\mathbb{N}, +)$$

is a semigroup because addition is closed on $\mathbb{N}$ and is associative.

2.2. Monoid

Definition

A **monoid** is a semigroup that contains an identity element.

Therefore,

$$\boxed{\text{Monoid} = \text{Semigroup} + \text{Identity}}$$

A monoid must satisfy:

1. Closure
2. Associativity
3. Identity

Example

Consider

$$(\mathbb{Z}, +).$$

Addition is closed:

$$a + b \in \mathbb{Z}.$$

Addition is associative:

$$(a + b) + c = a + (b + c).$$

The identity is 0:

$$a + 0 = 0 + a = a.$$

Therefore,

$$\boxed{(\mathbb{Z}, +) \text{ is a monoid.}}$$

2.3. Group

Definition

A **group** is a monoid in which every element has an inverse.

A group satisfies:

1. Closure
2. Associativity
3. Identity
4. Inverse

$$\boxed{\text{Group} = \text{Monoid} + \text{Inverse}}$$

Example

Consider

$$(\mathbb{Z}, +).$$

For every integer a, its additive inverse is $-a$, because

$$a + (-a) = 0.$$

Therefore,

$$\boxed{(\mathbb{Z}, +) \text{ is a group.}}$$

2.4. Abelian Group

Definition

A group G is called an **abelian group** if the operation is commutative. That is,

$$\boxed{a * b = b * a}$$

for all $a, b \in G$.

Example:

$$(\mathbb{Z}, +)$$

is an abelian group because

$$a + b = b + a.$$

3. Subgroups

Definition

Let G be a group. A non-empty subset $H \subseteq G$ is called a **subgroup** of G if H itself forms a group under the same operation as G.

We write

$$\boxed{H \leq G}.$$

3.1. Subgroup Test

For a multiplicative group:

$$\boxed{a, b \in H \Rightarrow ab^{-1} \in H}$$

For an additive group:

$$\boxed{a, b \in H \Rightarrow a - b \in H}$$

Example

Consider the set of even integers:

$$2\mathbb{Z} = \{\ldots, -4, -2, 0, 2, 4, \ldots\}.$$

We know that

$$(\mathbb{Z}, +)$$

is a group.
Take

$$a = 2m, \qquad b = 2n.$$

Then

$$a - b = 2m - 2n$$

$$= 2(m - n).$$

Since $m - n \in \mathbb{Z}$,

$$2(m - n) \in 2\mathbb{Z}.$$

Therefore,

$$\boxed{2\mathbb{Z} \leq (\mathbb{Z}, +)}.$$

3.2. PYQ: Intersection of Two Subgroups

Previous-Year Question

Question:
If H and K are subgroups of a group G, prove that

$$H \cap K$$

is also a subgroup of G.

Proof:
We use the subgroup test.
Since H and K are subgroups of G,

$$e \in H$$

and

$$e \in K.$$

Therefore,

$$e \in H \cap K.$$

Hence,

$$H \cap K \neq \varnothing.$$

Now let

$$a, b \in H \cap K.$$

Then

$$a, b \in H$$

and

$$a, b \in K.$$

Since H is a subgroup,

$$ab^{-1} \in H.$$

Since K is a subgroup,

$$ab^{-1} \in K.$$

Therefore,

$$ab^{-1} \in H \cap K.$$

Hence, by the subgroup test,

$$\boxed{H \cap K \leq G}.$$

Therefore, $H \cap K$ is a subgroup of G.

4. Homomorphism

Definition

A mapping

$$f : (A, *) \to (B, \circ)$$

is called a **homomorphism** if it preserves the operation.
That is,

$$\boxed{f(a * b) = f(a) \circ f(b)}$$

for every $a, b \in A$.

In simple words:

A homomorphism preserves the algebraic operation while mapping from one structure to another.

4.1. How to Prove a Function is a Homomorphism

Suppose

$$f : (A, *) \to (B, \circ).$$

To prove that f is a homomorphism:

1. Take arbitrary $a, b \in A$.
2. Find $f(a * b)$.
3. Find $f(a) \circ f(b)$.
4. Compare the two.

If

$$f(a * b) = f(a) \circ f(b),$$

then f is a homomorphism.

4.2. Example

Consider

$$f : \mathbb{Z} \to \mathbb{Z}$$

defined by

$$f(x) = 3x.$$

We check:

$$f(a + b) = 3(a + b)$$

$$= 3a + 3b.$$

Also,

$$f(a) + f(b) = 3a + 3b.$$

Therefore,

$$f(a + b) = f(a) + f(b).$$

Hence,

$$\boxed{f \text{ is a homomorphism.}}$$

5. Isomorphism

Definition

An **isomorphism** is a bijective homomorphism.
Therefore, an isomorphism must be:

1. A homomorphism
2. One-to-one
3. Onto

Homomorphism	**Isomorphism**
Preserves operation	Preserves operation
Need not be one-to-one	Must be one-to-one
Need not be onto	Must be onto
May lose some information	Preserves complete structure

Important

Isomorphism = Bijective Homomorphism

5.1. PYQ: Homomorphism from $\mathbb{Z}$ to $\mathbb{Z}_4$

Previous-Year Question

Question:
Define homomorphic functions. Consider

$$(\mathbb{Z}, +)$$

and

$$(\mathbb{Z}_4, +_4).$$

Define

$$g : \mathbb{Z} \to \mathbb{Z}_4$$

by

$$g(a) = a \pmod 4.$$

Show that g is a homomorphism.

Solution:
For $a, b \in \mathbb{Z}$,

$$g(a+b) = (a+b) \pmod 4.$$

Using the property of modulo addition,

$$(a+b) \pmod 4 = [(a \pmod 4) + (b \pmod 4)] \pmod 4.$$

But

$$g(a) = a \pmod 4$$

and

$$g(b) = b \pmod 4.$$

Therefore,

$$g(a+b) = g(a) +_4 g(b).$$

Hence,

$$\boxed{g(a+b) = g(a) +_4 g(b)}.$$

Therefore,

$$\boxed{g \text{ is a homomorphism.}}$$

6. Groups: Important Example

Previous-Year Question

PYQ:
Let $\mathbb{Z}$ be the set of integers. Define $*$ in $\mathbb{Z}$ by

$$a * b = a + b + 2$$

for all $a, b \in \mathbb{Z}$.
Verify whether

$$(\mathbb{Z}, *)$$

is a group.

6.1. Step 1: Closure

Take

$$a, b \in \mathbb{Z}.$$

Then

$$a * b = a + b + 2.$$

Since the sum of integers is an integer,

$$a + b + 2 \in \mathbb{Z}.$$

Therefore, closure holds.

$$\boxed{\text{Closure: Yes}}$$

6.2. Step 2: Associativity

Consider

$$(a * b) * c.$$

Since

$$a * b = a + b + 2,$$

we get

$$(a * b) * c = (a + b + 2) + c + 2$$

$$= a + b + c + 4.$$

Now consider

$$a * (b * c).$$

Since

$$b * c = b + c + 2,$$

we get

$$a * (b * c) = a + (b + c + 2) + 2$$

$$= a + b + c + 4.$$

Therefore,

$$(a * b) * c = a * (b * c).$$

Hence,

$$\boxed{\text{Associativity: Yes}}$$

6.3. Step 3: Identity Element

Let the identity element be e.

We require

$$a * e = a.$$

Therefore,

$$a + e + 2 = a.$$

Subtract a from both sides:

$$e + 2 = 0.$$

Hence,

$$\boxed{e = -2}.$$

Check:

$$a * (-2) = a + (-2) + 2 = a.$$

Also,

$$(-2) * a = -2 + a + 2 = a.$$

Therefore,

$$\boxed{-2 \text{ is the identity element.}}$$

6.4. Step 4: Inverse Element

Let the inverse of a be b.

We require

$$a * b = e.$$

Since

$$e = -2,$$

we have

$$a + b + 2 = -2.$$

Therefore,

$$a + b = -4.$$

Hence,

$$\boxed{b = -4 - a}.$$

Since $a \in \mathbb{Z}$,

$$-4 - a \in \mathbb{Z}.$$

Therefore, every element has an inverse.

6.5. Step 5: Conclusion

All group properties are satisfied:

Property	Result
Closure	Yes
Associativity	Yes
Identity	-2
Inverse	$-4 - a$

Therefore,

$$\boxed{(\mathbb{Z}, *) \text{ is a group.}}$$

Moreover,

$$a * b = a + b + 2 = b + a + 2 = b * a.$$

Therefore, the operation is also commutative.
Hence,

$$\boxed{(\mathbb{Z}, *) \text{ is an Abelian group.}}$$

7. Lattices

7.1. Partial Order

A relation $\leq$ on a set P is called a **partial order** if it satisfies:

1. Reflexive
2. Antisymmetric
3. Transitive

Therefore,

$$\boxed{\text{Partial Order} = \text{Reflexive} + \text{Antisymmetric} + \text{Transitive}}$$

A set together with a partial order is called a **poset**.

$$\boxed{(P, \leq)}$$

is called a partially ordered set or poset.

7.2. Lattice Definition

Definition

A **lattice** is a poset in which every pair of elements has:

1. A greatest lower bound (GLB), called the **meet**
2. A least upper bound (LUB), called the **join**

The meet is represented by

$$\boxed{a \wedge b}$$

and the join is represented by

$$\boxed{a \vee b}.$$

Therefore,

$$\boxed{\text{Lattice} = \text{Poset} + \text{GLB and LUB for every pair}}$$

7.3. Understanding Lower Bound and Upper Bound

Suppose we have two elements a and b.
A **lower bound** is an element that is less than or equal to both.
A **greatest lower bound (GLB)** is the largest among all lower bounds.
An **upper bound** is an element that is greater than or equal to both.
A **least upper bound (LUB)** is the smallest among all upper bounds.

Important

Memory Trick

$$\boxed{\text{Meet} = \text{GLB}}$$

$$\boxed{\text{Join} = \text{LUB}}$$

Think:

$$\boxed{\wedge = \text{Meet} = \text{Lower}}$$

$$\boxed{\vee = \text{Join} = \text{Upper}}$$

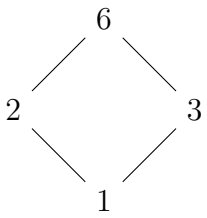
7.4. Example of a Lattice

Consider

$$L = \{1, 2, 3, 6\}$$

under the divisibility relation.

The Hasse diagram is:



Consider the elements 2 and 3.
The lower bounds are:

$$\{1\}.$$

Therefore,

$$GLB(2,3) = 1.$$

Thus,

$$\boxed{2 \wedge 3 = 1}.$$

The upper bounds are:

$$\{6\}.$$

Therefore,

$$LUB(2,3) = 6.$$

Thus,

$$\boxed{2 \vee 3 = 6}.$$

For this example,

$$2 \wedge 3 = \gcd(2,3) = 1$$

and

$$2 \vee 3 = \operatorname{lcm}(2,3) = 6.$$

8. Basic Laws of Lattices

8.1. 1. Idempotent Law

$$\boxed{a \vee a = a}$$

and

$$\boxed{a \wedge a = a}.$$

8.2. 2. Commutative Law

$$\boxed{a \vee b = b \vee a}$$

and

$$\boxed{a \wedge b = b \wedge a}.$$

8.3. 3. Associative Law

$$\boxed{(a \vee b) \vee c = a \vee (b \vee c)}$$

and

$$\boxed{(a \wedge b) \wedge c = a \wedge (b \wedge c)}.$$

8.4. 4. Absorption Law

$$\boxed{a \vee (a \wedge b) = a}$$

and

$$\boxed{a \wedge (a \vee b) = a}.$$

9. Types of Lattices

Previous-Year Question

PYQ: Describe different types of lattices.

The important types of lattices are:

1. Bounded lattice
2. Complemented lattice
3. Distributive lattice
4. Complete lattice

9.1. 1. Bounded Lattice

Definition

A lattice is called a **bounded lattice** if it contains:

- A least element, usually denoted by 0
- A greatest element, usually denoted by 1

such that

$$\boxed{0 \leq a \leq 1}$$

for every $a \in L$.

In a bounded lattice,

$$0 \wedge a = 0$$

and

$$1 \vee a = 1.$$

Also,

$$0 \vee a = a$$

and

$$1 \wedge a = a.$$

9.2. 2. Complemented Lattice

Definition

A bounded lattice is called a **complemented lattice** if every element a has a complement a' such that

$$\boxed{a \vee a' = 1}$$

and

$$\boxed{a \wedge a' = 0}.$$

In simple words:

Every element must have another element which combines with it to give the greatest element through join and the least element through meet.

9.3. 3. Distributive Lattice

Definition

A lattice is called a **distributive lattice** if the distributive laws are satisfied.

$$\boxed{a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)}$$

and

$$\boxed{a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)}.$$

There are two distributive laws:

$$\boxed{\wedge \text{ distributes over } \vee}$$

and

$$\boxed{\vee \text{ distributes over } \wedge}.$$

9.4. 4. Complete Lattice

Definition

A lattice is called a **complete lattice** if every subset of the lattice has both:

- a greatest lower bound (GLB)
- a least upper bound (LUB)

In simple words:

Every subset must have a meet and a join.

9.5. Quick Comparison of Types of Lattices

Type	Meaning
Bounded	Has a least element 0 and greatest element 1.
Complemented	Every element a has a complement a' satisfying $a \vee a' = 1$ and $a \wedge a' = 0$.
Distributive	Satisfies the two distributive laws involving meet and join.
Complete	Every subset has both GLB and LUB.

10. Important Theorems

10.1. Theorem 1: Uniqueness of Identity

Statement:

The identity element of a group is unique.

Proof:

Suppose e and e' are both identity elements.
Since e is an identity,

$$e * e' = e'.$$

Since e' is an identity,

$$e * e' = e.$$

Therefore,

$$e = e'.$$

Hence, the identity element is unique.

$$\boxed{\therefore \text{ Identity is unique.}}$$

10.2. Theorem 2: Uniqueness of Inverse

Statement:
Every element in a group has a unique inverse.
Proof:
Suppose b and c are both inverses of a.
Then

$$a * b = b * a = e$$

and

$$a * c = c * a = e.$$

Now,

$$\begin{aligned} b &= b * e \\ &= b * (a * c) \\ &= (b * a) * c \\ &= e * c \\ &= c. \end{aligned}$$

Therefore,

$$\boxed{b = c}.$$

Hence, the inverse is unique.

10.3. Theorem 3: Inverse of a Product

For elements a, b in a group,

$$\boxed{(ab)^{-1} = b^{-1}a^{-1}}.$$

This is called the **reversal rule**.

To verify,

$$(ab)(b^{-1}a^{-1})$$

$$= a(bb^{-1})a^{-1}$$

$$= a(e)a^{-1}$$

$$= aa^{-1}$$

$$= e.$$

Similarly,

$$(b^{-1}a^{-1})(ab) = e.$$

Therefore,

$$\boxed{(ab)^{-1} = b^{-1}a^{-1}}.$$

Important

Memory Trick:
When finding the inverse of a product, **reverse the order**.

$$\boxed{(ab)^{-1} = b^{-1}a^{-1}}$$

10.4. Theorem 4: Subgroup of an Abelian Group

Statement:
Every subgroup of an abelian group is also abelian.
Proof:
Let

$$H \leq G$$

and suppose G is abelian.
Take any

$$a, b \in H.$$

Since

$$H \subseteq G,$$

we have

$$a, b \in G.$$

Since G is abelian,

$$ab = ba.$$

Therefore, the operation in H is also commutative.
Hence,

$$\boxed{H \text{ is abelian.}}$$

11. Previous-Year Questions: Quick Revision

No.	Question	Main Topic
1	Define semigroup with example.	Semigroup
2	Define a lattice.	Lattice
3	Prove that $(\mathbb{Z}, +)$ is a monoid.	Monoid
4	If H and K are subgroups of G, prove that $H \cap K$ is also a subgroup of G.	Subgroup
5	Let $a * b = a + b + 2$. Verify whether $(\mathbb{Z}, *)$ is a group.	Group
6	Define homomorphic functions and show that $g : \mathbb{Z} \to \mathbb{Z}_4$, $g(a) = a \pmod 4$, is a homomorphism.	Homomorphism
7	Describe different types of lattices.	Types of lattices

12. Exam Writing Strategy

12.1. If Asked: Define Semigroup

Write:

> A semigroup is a non-empty set together with a binary operation which is closed and associative.

Then give one example.

$$(\mathbb{N}, +)$$

is a semigroup.

12.2. If Asked: Prove $(\mathbb{Z}, +)$ is a Monoid

Write the properties in this order:

1. Closure
2. Associativity
3. Identity

Then conclude:

$$\boxed{(\mathbb{Z}, +) \text{ is a monoid.}}$$

Note: Since every integer also has an additive inverse, $(\mathbb{Z}, +)$ is actually a group and an abelian group.

12.3. If Asked: Verify Whether an Operation Forms a Group

Always check in this order:

$$\boxed{\text{Closure} \rightarrow \text{Associativity} \rightarrow \text{Identity} \rightarrow \text{Inverse}}$$

If all four are satisfied, it is a group.
If the operation is also commutative, it is an abelian group.

12.4. If Asked: Prove Homomorphism

Remember:

$$\boxed{f(a * b) = f(a) \circ f(b)}$$

Take arbitrary a, b, calculate both sides and show they are equal.

12.5. If Asked: Define Lattice

Write:

> A lattice is a poset in which every pair of elements has a greatest lower bound (GLB), called meet, and a least upper bound (LUB), called join.

Then write:

$$\boxed{a \wedge b = \text{GLB}}$$

$$\boxed{a \vee b = \text{LUB}}$$

13. One-Page Quick Revision

Concept	**Key Point**
Binary operation	$* : A \times A \to A$
Semigroup	Closure + Associativity
Monoid	Semigroup + Identity
Group	Monoid + Inverse
Abelian Group	Group + Commutativity
Subgroup	Subset which itself forms a group
Homomorphism	Preserves operation: $f(a * b) = f(a) \circ f(b)$
Isomorphism	Bijective homomorphism
Poset	Reflexive + Antisymmetric + Transitive
Lattice	Every pair has GLB and LUB
Meet	GLB: $a \wedge b$
Join	LUB: $a \vee b$
Bounded lattice	Has least element 0 and greatest element 1
Complemented lattice	$a \vee a' = 1, \quad a \wedge a' = 0$
Distributive lattice	Satisfies distributive laws
Complete lattice	Every subset has GLB and LUB

14. Memory Shortcuts

Important

1. Algebraic Structures

$$\boxed{\text{Semigroup} \to \text{Monoid} \to \text{Group} \to \text{Abelian Group}}$$

$$\boxed{CA \to CAI \to CAII \to CAIIC}$$

where

$C = \text{Closure}, \quad A = \text{Associativity}, \quad I = \text{Identity/Inverse}, \quad C = \text{Commutativity}.$

2. Homomorphism

$$\boxed{\text{Operation is preserved}}$$

$$f(a * b) = f(a) \circ f(b).$$

3. Isomorphism

$$\boxed{\text{Isomorphism} = \text{Homomorphism} + \text{One-to-one} + \text{Onto}}$$

4. Lattice

$$\boxed{\text{Meet} = \text{GLB}}$$

$$\boxed{\text{Join} = \text{LUB}}$$